

Dimension reduction by propensity score

Can we use a propensity score instead of covariates?

Consider the regression model¹

$$y = \beta_0 + \beta_d d + \beta_x x + \epsilon \quad (1)$$

where y is the outcome or dependent variable, d is a dummy variable representing group/class membership, and x is a single covariate. Here d and x are the independent or predictor variables. To simplify the math we will assume that x is centered.

We'll use the Frisch-Waugh-Lovell (FWL) Theorem to estimate β_d .

The FWL theorem states that any predictor's regression coefficient in a multivariate model is equivalent to the regression coefficient estimated from a bivariate model in which the residualised outcome is regressed on the residualized component of the predictor; where the residuals are taken from models regressing the outcome **and** the predictor on all other predictors in the multivariate regression (separately).

First residual - the propensity score:

The propensity score predicts the dummy variable using the covariates, i.e the regression:

$$d = \beta_x^{(d)} x + \epsilon^{(d)}$$

where $\beta_x^{(d)} x = x \frac{\text{cov}(x,d)}{\text{var}(x)}$. We define the propensity score as $P(x) = x \frac{\text{cov}(x,d)}{\text{var}(x)}$.

Thus $d - P(x)$ is the first leg of the FWL Theorem applied to Equation 1.

¹this can be generalized to multiple covariates.

Second residual:

The second leg of the FWL Theorem starts with the regression:

$$y = \beta_x^{(y)} x + \epsilon^{(y)} \quad (2)$$

where $\beta_x^{(y)} = \frac{\text{cov}(x, y)}{\text{var}(x)}$, and the second leg of the FWL Theorem is $y - x \frac{\text{cov}(x, y)}{\text{var}(x)}$.

Can we regress on $P(x)$ instead of x in Equation 2?

For our second leg, instead of Equation 2 can we use the following regression?

$$y = \beta_{P(x)}^{(y)} P(x) + \epsilon^{(y)} \quad (3)$$

We can write Equation 3 as $y = P(x) \frac{\text{cov}(P(x), y)}{\text{var}(P(x))} + \epsilon^{(y)}$, and then note that:

- $P(x) \frac{\text{cov}(P(x), y)}{\text{var}(P(x))} = P(x) (P(x)^\top P(x))^{-1} \text{cov}(P(x), y)$, and

$$P(x) (P(x)^\top P(x))^{-1} \text{cov}(P(x), y) = x \frac{\text{cov}(d, x)}{\text{var}(x)} \times \frac{\text{var}(x)}{(\text{cov}(d, x))^2} \times \frac{\text{cov}(d, x)}{\text{var}(x)} \text{cov}(x, y)$$

- since $P(x)^\top P(x) = \frac{\text{cov}(d, x)}{\text{var}(x)} x^\top x \frac{\text{cov}(d, x)}{\text{var}(x)} = \frac{(\text{cov}(d, x))^2}{\text{var}(x)}$, and
- $\text{cov}(P(x), y) = \text{cov}(\frac{\text{cov}(d, x)}{\text{var}(x)} x, y) = \frac{\text{cov}(d, x)}{\text{var}(x)} \text{cov}(x, y)$
- so $P(x) \frac{\text{cov}(P(x), y)}{\text{var}(P(x))} = x \frac{\text{cov}(x, y)}{\text{var}(x)}$ and they are exchangeable.

It follows that in the regression $y = \beta_0 + \beta_d d + \beta_x x + \epsilon$, with the propensity score $P(x) = \frac{\text{cov}(d, x)}{\text{var}(x)} x$, then per the Frisch-Waugh theorem the coefficient β_d is equal to the coefficient from regressing

$$y - \frac{\text{cov}(P(x), y)}{\text{var}(P(x))} P(x) \text{ on } d - P(x)$$

With multiple covariates:

- For a covariate matrix X the propensity score is written as $P(X) = X \frac{X^\top d}{X^\top X} = X(X^\top X)^{-1} X^\top d$

- For $P(x) \frac{\text{cov}(P(x), y)}{\text{var}(P(x))}$

– $\text{var}(P(x)) = P(x)^\top P(x)$ which is

$$(X(X^\top X)^{-1} X^\top d)^\top \times X(X X^\top)^{-1} X^\top d = d^\top X(X^\top X)^{-1} X^\top \times X(X^\top X)^{-1} X^\top d$$

which simplifies to $d^\top X(X X^\top)^{-1} X^\top d$

- $(P(x)^\top P(x))^{-1}$ is $(d^\top X(X X^\top)^{-1} X^\top d)^{-1}$ which is $(X^\top d)^{-1} (X X^\top)(d^\top X)^{-1}$
- $\text{cov}(P(x), y)$ is $P(x)^\top y$ which is $(X(X^\top X)^{-1} X^\top d)^\top y = d^\top X(X X^\top)^{-1} X^\top y$, so
- $\frac{\text{cov}(P(x), y)}{\text{var}(P(x))}$ is $(X^\top d)^{-1} (X X^\top)(d^\top X)^{-1} d^\top X(X X^\top)^{-1} X^\top y = (X^\top d)^{-1} X^\top y$, and finally
- $P(x) \frac{\text{cov}(P(x), y)}{\text{var}(P(x))} = X(X^\top X)^{-1} X^\top d (X^\top d)^{-1} X^\top y$ which is $X(X^\top X)^{-1} X^\top y = X \frac{\text{cov}(X, y)}{\text{var}(X)}$